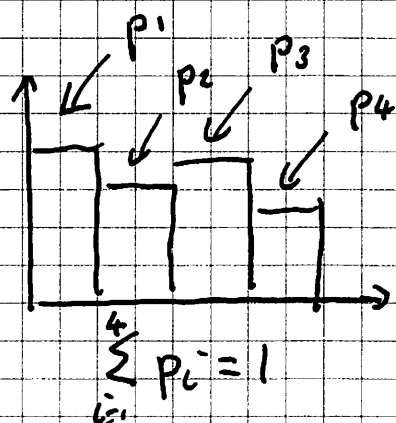


Edinburgh Talk - Jaynes World

①

Histogram



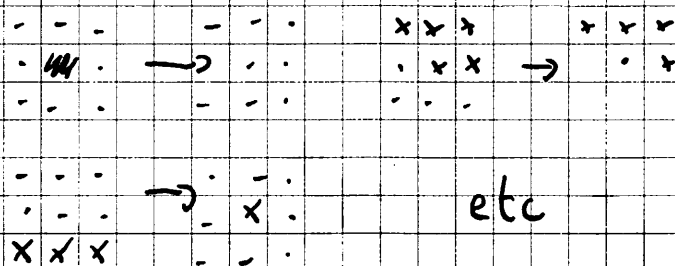
Entropy defined as

$$S(\theta) = - \sum_x p(x|\theta) \log p(x|\theta)$$

Cellular Automata

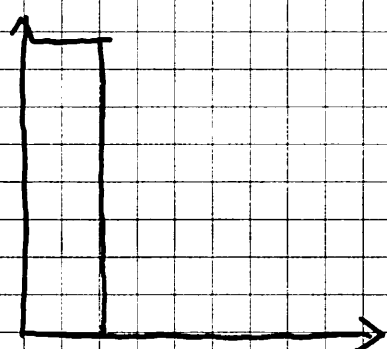
Conway's Life

Wolfram's Rules

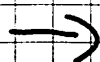
Simple rules that
lead to complex
behaviour

Turing Complete

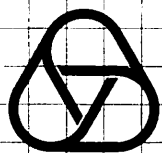
Jaynes World - Steepest ascent in Entropy



Low Entropy



High Entropy



Game Rule

(2)

$$\theta_{t+1} = \theta_t + \mu \frac{ds(\theta)}{d\theta}$$

Deterministic Rule

Entropy Compact Manifold $0 \rightarrow N$
(No Singularities)

Stochastic System

① Origin. Multiple Global Maxima

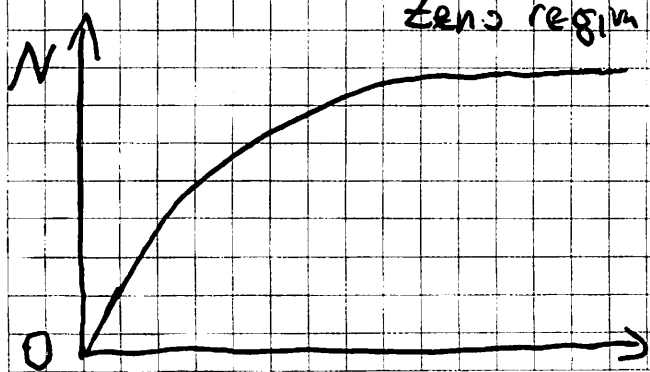
Conjecture: Due to Steepest Ascent

Your Game always appears to start at a minimal entropy configuration, θ_0

② Valid Game: If you return to Origin on Gradient Path ^{from θ_0} and ^{to θ_0} cannot climb back to θ_t from θ_0 . The game θ_t is invalid

③ End Game: Max entropy is unique

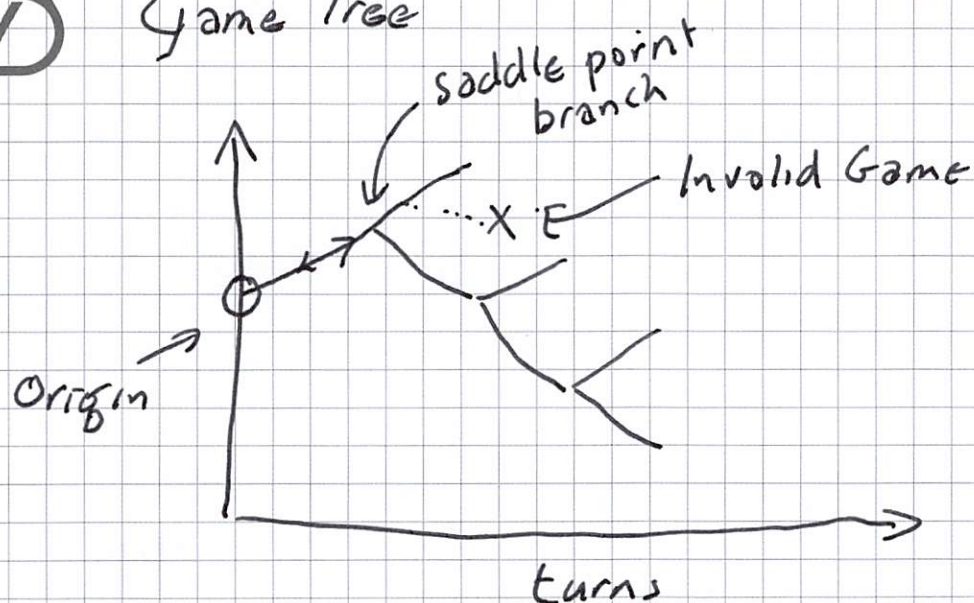
Zeno's Paradox in game turns.





③

Game Tree

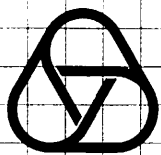


Morse

① Complexity of a path defined as number of saddle points passed Theory

② Complexity of a Game is expected number of saddle points passed averaged over all paths.

③ ~~Simple game~~ ~~is one where all paths lead to same end game~~



④

Game Configuration

Life has N cells on a grid with
co-ordinates X in $D = 2$ dimensions

Our game - N cells on a grid with
 N dimensions, sample

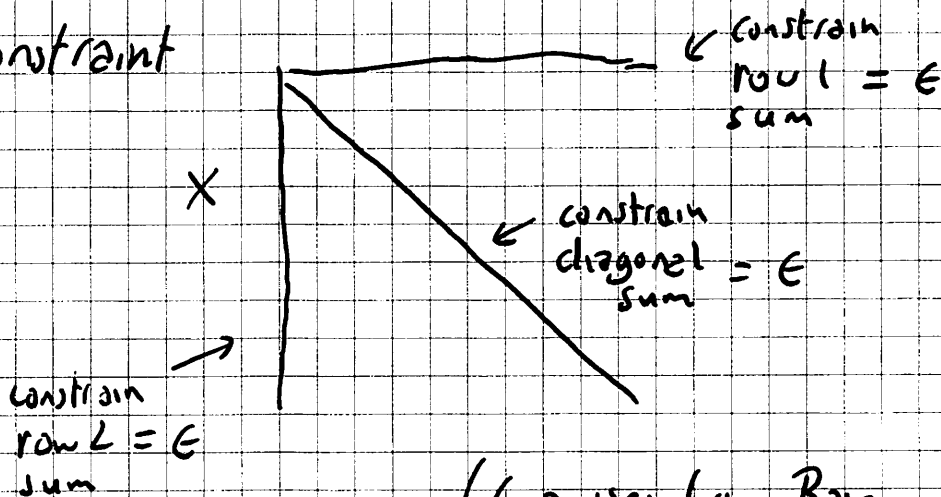
$$x \sim \mathcal{N}(0, \sigma^2)$$

$$\Rightarrow X \in \mathbb{R}^{N \times N}$$

Minimal
configuration.

Permutation constraints on Rows & Columns?

Bracing Constraint
(Spatial)

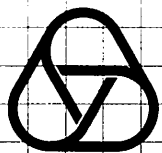


(ϵ arises from Base
measure in differential
entropy)

Temporal Constraint

$$\text{Constrain } X_{00} - X_1 = \epsilon$$

Tautology - Teleology Alert!



Kronecker Sum

⑤

Kronecker Product

$$K \otimes I_2 = \begin{pmatrix} K & 0 \\ 0 & K \end{pmatrix}$$

$$I \otimes K = \begin{pmatrix} k_{11} & 0 & k_{12} & 0 \\ 0 & k_{11} & 0 & k_{12} \\ k_{21} & 0 & k_{22} & 0 \\ 0 & k_{21} & 0 & k_{22} \end{pmatrix}$$

Kronecker Sum $C_1 \oplus C_2 = C_1 \otimes I + I \otimes C_2 = C$

$$C_1 = U_1 \Lambda^2 U_1^T \quad C_2 = U_2 L^2 U_2^T$$

Eigenvalues

$$U_1 \otimes U_2 (\Lambda^2 \oplus L^2) U_1^T \otimes U_2^T$$

so eigen values are

$$\begin{matrix} \lambda_1^2 + l_1^2 \\ \lambda_2^2 + l_1^2 \\ \vdots \\ \lambda_N^2 + l_1^2 \\ \lambda_1^2 + l_2^2 \\ \vdots \\ \lambda_N^2 + l_2^2 \\ \vdots \\ \lambda_1^2 + l_N^2 \\ \vdots \\ \lambda_N^2 + l_N^2 \end{matrix}$$

$$\text{Entropy } S(\theta) = \frac{1}{2} \sum_i \log(e \tau (\lambda_i^2 + l_i^2))$$

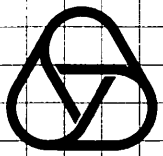
$$\frac{dS(\theta)}{d\lambda_i} = \frac{\lambda_i}{(\lambda_i^2 + l_i^2)}$$

$$\frac{d^2 S(\theta)}{d\lambda_i^2} = \frac{-2\lambda_i^2}{(\lambda_i^2 + l_i^2)^2}$$

$$\frac{d^2 S(\theta)}{d\lambda_i d l_i} = \frac{-2\lambda_i l_i}{(\lambda_i^2 + l_i^2)^2} \quad H = -2JJ^T$$

Optimal Gradient is $H^{-1}J^T = J^{-T}$

But we travel in J^T !



Game Board Development

⑥

Three spatial constraints implies

Three $\lambda_i^2 \in \mathbb{R}_i \in \mathbb{B}_i$

One temporal constraint implies

One $\lambda_t^2 \in \mathbb{R}_t \in \mathbb{B}$

Assume all others are $\gg \epsilon$ and scaled $\mathcal{O}(1)$

Eigen values are ~~natural~~ ^{natural} parameters

Gradient step in eigenvalues is unnatural
as it is orthogonal to the natural step

For $\frac{\lambda_i}{\lambda_i^2 + \lambda_t^2}$ Gradient is $\frac{1}{\sqrt{\epsilon}}$ if $i \text{ or } t \in \mathbb{B}$
and $\mathcal{O}(1)$ if $i \text{ or } t \notin \mathbb{B}$

This implies first step three spatial
eigen values

"explode/inflate"

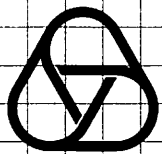
One temporal eigen value "explodes/inflates"

Other directions get "stuck" at $\mathcal{O}(1)$

Board set-up - 3 large spatial dimensions. 1 temporal

Constrain temporal to fixed value to ensure

spatial are "measurable" - T-T warning!



⑦

Next Steps

① Characterise constraints properly
(T - τ problem)

② Specify cell states & rules

- Game or Games analysis (independent)
- Cell states operate at small length/time scales (small λ, ℓ) relative to board

③ Path Integrals to characterise "physics"

What does an information action principle look like? B. Ry. Frieden's GPI

④ Implications of rules at different scales

- Thermodynamics of info
- Information engines
- Information reservoirs

⑤ Implications for implicit regularisation
Kronecker sum + Kronecker product and
 $\Theta_i = \log(\ell_i)$ parameterisation implies

composition of $\log \log \frac{1}{1 + e^{\sigma_i - \sigma_j}}$

terms in objective.

Implications for deep NNs
& entropy descent.